# Branson Tay

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## WORK EXPERIENCE

**Jaguar TCS Racing** **Aug. 2024 – Aug. 2025**
*Junior Strategy & Software Engineer* *Kidlington, United Kingdom*

**Stack**: Python, MATLAB, ReactJS, Typescript, CSS, MongoDB, Azure Pipelines, Docker, VBA

- Pioneered and deployed a data processing python-pipeline converting Timing Feed data in noSQL databases to processed ATLAS telemetry channels, ensuring reliability via error handling, testing, and pipeline dev.
- Wrote a custom library to extract and write car data to ATLAS format in Python using MAT Atlas DLLs letting the Performance group engage in ML/AI projects such as virtual sensor data, or large aggregations of car data.
- Assisted in live race-weekend troubleshooting for a range of race-critical tools, improving software reliability.
- Optimised GPS node lookup with spatial data structures (KD-trees) for fast distance mapping and tow analysis.
- Develop full-stack ReactJS apps visualising data critical to race strategy with live telemetry, tyre & timing data.
- Process pre-race battery simulations and simulator data results to monitor battery temperatures and ensure legality, correlating with track data to ensure accuracy in our models and predict race temperature profiles.
- Logged and streamed live telemetry from WinTax using Visual Basic Scripts to a TCP socket via REST APIs.
- Perform qualitative competitor radio analysis (energy management, pit-stops, strategy, battery temperature management, vehicle setup and tyre run-plans) delivering concise per-session summaries to strategists.
- Support live traffic management in Berlin Rookie Test using RaceWatch, monitoring GPS and timing feed data.
- Analysed driver telemetry in ATLAS to optimise energy management and performance to gain race pace.

**Republic of Singapore Navy** **Nov. 2020 – Jul. 2022**
*Marine Systems Specialist Technician* *Singapore, Singapore*

- Trained on Singapore Navy's largest marine vessel, on maintenance and wartime awareness of ship systems.
- Led remote war-time firefighting simulation aboard ship engine operations room to assign appropriate fixes.
    - Analysed live sensor data in core ship systems from telemetry displays under high-pressure situations.
- Contributed to 50 pre- and post-mission debriefs, highlighting potential improvements for team efficiency.

**Travelindr** **Oct. 2019 – Mar. 2020**
*Co-founder & CEO* *Singapore, Singapore*

- Founded Travelindr, a group travel-itinerary generator, in a small team implementing Agile practices.
- Oversaw core strategies, product roadmap, data & competitor analysis, talent acquisition & market research.
- Pitched to investors, achieved Minimal-Viable Product status, incubated in esteemed multi-national company.

## EDUCATION

**University of Leeds** **Sep 2022. – Jul. 2026**
*BSc Computer Science with Year in Industry*
*Leeds, UK*

- **Formula Student Performance & Simulation Sub-team Lead**: Lead development of simulation-driven design workflow using IPG and Python quasi-static steady-state lap time sim to guide vehicle design decisions.
- **Proposed Final Year Project**: Building a race strategy simulation platform using multi-agent reinforcement learning with 20 autonomous agents. Models key race dynamics including tyre wear, safety car likelihood, overtaking difficulty, and energy deployment, relying on policy convergence to determine ideal strategies.

**Ngee Ann Polytechnic** **Jan. 2017 - Mar. 2020**
*Diploma in Mechanical Engineering* *Singapore, Singapore*

- Developed a gearbox meeting specific load, speed, and dimensional requirements for efficient material handling.
- Utilised AutoCAD CAD software to fabricate gearbox given specific load, speed, and dimensional requirements
- Recognized with the Good Progress Award and the School of Engineering Merit Award.

## TECHNICAL PROJECTS

**Formula Student Lap Time Simulator |Python, Vehicle Dynamics, Control Theory, Git**

- Developed quasi-static steady-state point-mass and bicycle model lap time simulators using TTC tyre data, enabling sensitivity analysis of design parameters such as track width, wheelbase, and chassis stiffness.

**2D CFD Formula Student Radiator Optimisation Simulator |Rust, Numerical Methods**

- Developed a 2D incompressible fluid solver analysing radiator angle effects on cooling, pressure drop and drag.
- Automated batch simulations and visualisations to optimise radiator design, leveraging parallel processing.

**Formula Student ICE Data Acquisition System|C++, Git, Arduino**

- Aggregated data from GPS module to Arduino Teensy, visualising live data to the driver dash and over radio frequency for multiple departments to better quantify and enhance vehicle performance.
- Honed proficiency in collaborative software engineering tools like Git and conducting code reviews.

**Formula Student EV Battery Management System|C++, CAN Bus protocol**

- Spearheaded EV emulation of Battery Management System temperature sensor extension embedded system.
- Architected an expansion module using the CAN bus protocol for robust and efficient data transfer.
- Designed microprocessor framework to process battery module voltages to tangible data in C++.

**Pathfinder Visualiser| ReactJS, Flask, HTML, CSS, JS**

- Programmed a Full-Stack Web Application with Flask serving as a REST API and React as the frontend.
- Employed Dijkstra's algorithm and A* algorithm to visually demonstrate the speed difference between them.

## ADDITIONAL EXPERIENCE

**E-commerce Shopify Store Building | Facebook Ads Manager, Adobe Photoshop, Adobe Premiere Pro**

- Developed and marketed multiple e-commerce shop brands by creating catchy, attention-grabbing creatives.
- Drove user attention by utilising good design principles like obvious action buttons, and consistent elements.
    - Increased conversions in by 10% by adapting "add-to-cart" button colour based on cursor heatmaps.
- Constructed efficient sales funnel, extracting data insights, and targeting trends, fully utilising data analysis tools.

**Startup Partnership Intern***,* Global Entrepreneurial Internship Scholar

- Pioneered Voucher System project integrating cross-department efforts within a strict one-month window.
    - Achieved a 10% surge in the 30,000 annual customer base nationally.
- Independently secured over 10 successful strategic partnerships through presentations with senior executives.

## TECHNICAL SKILLS, SOFT SKILLS & CERTIFICATIONS

**Languages**: Python, Matlab, HTML, CSS, Typescript, VBS, MongoQL, Rust, Java, C++, C.
**Frameworks and Libraries**: NumPy, SciPy, Bootstrap, Matplotlib, ReactJS, NodeJS, MongoDB, MySQL, Flask.
**Tools**: Docker, Git, Git Bash, Copilot, Azure DevOps, Azure Pipelines, Visual Studio, Microsoft Suite, ATLAS.

**Soft Skills**: Collaborative | Time-management | Curious | Adaptable | Multiperspectivity

**Certifications & Awards**

- Completed Harvard's introductory course to computer science, CS50.
- Runner-up in TRIVE Global Entrepreneurial Internship Program Startup Challenge.

## HOBBIES & PERSONAL INTERESTS

- Engineering: Avid viewer of motorsport and airplane engineering videos|Karting.
- Stress-relieving: Padel|Table-tennis|Badminton|Cooking|Painting|Running.
- Enriching: Reading fiction & non-fiction books|Podcasts & videos on behavioural science.